Bartlett adjustment for random effects meta-analysis of event outcomes with small samples

Haben Michael[1]

[1]University of Massachusetts, Amherst, MA (hmichael@math.umass.edu)

Abstract. ... Keywords: Meta-analysis, Bartlett correction, Generalized linear mixed models

1. introduction: (skip generic metaanalysis material). There is currently underway a move toward generalized mixed models in carrying out meta-analyses of event outcomes, particularly in situations where sample sizes are small or events are rare. GLMMs avoid many of the dubious assumptions of the main competing approach, inverse-variance weighting, such as gaussianity of the effects and perfect knowledge of standard deviations. However, inferences under GLMMs are still based on an asymptotic chi-squared approximation. Therefore, as in the case of inverse-variance weighted estimation, the conclusions of smaller meta-analyses are suspect, while smaller meta-analyses are where use of GLMMs is often recommended. A classical remedy is Bartlett correction, which leverages modelling assumptions to improve the efficiency of the testing procedure. When the modelling assumptions are met, the imrpovement is typically a jump in the rate of asymptotic convergence from $1/n$ to $1/n^2$. [[the key modelling assumption in this setting is the gaussianity of the random effect.]]

While Bartlett corrections have long been applied to generalized linear models [[cite]], there do not appear to be extensions to even the simple GLMMs that are used in meta-analysis. The reason for the reluctance appears to be computational difficulty [[cite hanada]]. Unlike in the case of GLMs, closed-form expressions are unavailable for Bartlett corrections for GLMMs. We argue that with careful design these difficulties can be overcome and consumer-grade hardware is sufficient for practical implementation.

In the next section we summarize 4 event data meta-analysis models, following [[Stijnen's]] exposition. In Section ((ref)) we describe a practical approach to estimation of the Bartlett correction applicfable to these 4 models. In Section ((ref)) we examine the effectiveness and robustness of the corrected test, finding ... . In Section ((ref)) we re-visit the data analyses of ((stijnen)), finding ... .

2. background.

2a. [[notation for events, trials.]] We consider the random effects model for meta-analysis. In this model, a Gaussian random effects models the effect of the individual studies,

$$\theta_i = \mu_0 + \tau_0 Z, Z \sim N(0,1)$$

the target of inference in the meta-analysis is μ_0, with τ_0^2 a nuisance parameter. the conditional distribution of the data given the random effect depends on the type of analysis. following stijnen we consider 4 types of analyses: single arm and two arm trials, proportions and incidence/rates.

1. Single arm, proportions. Each study reports the number of events Y_i and total number of trials m_i. The statistical model is binomial

$$f(y_i \mid Z) = \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i}$$
$$p = \text{expit}(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the log-odds of an event. The number of trials m_i is treated as non-random.

2. Two arms, proportions. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the total number of trials $m_i^{(T)}$ and $m_i^{(C)}$. The statistical model conditions on the total number of events $S_i = Y_i^{(T)} + Y_i^{(C)}$, under which $Y_i^{(T)}$ follows Fisher's non-central hypergeometric distribution

$$f(y_i^{(T)} \mid Z_i, S_i) = \frac{\binom{m_i^{(T)}}{y_i^{(T)}}\binom{m_i^{(C)}}{S_i - y_i^{(T)}} \exp(\theta_i y_i^{(T)})}{\sum_{\max(0, S_i - m_i^{(C)})}^{\min(S_i, m_i^{(T)})} \binom{m_i^{(T)}}{y}\binom{m_i^{(C)}}{S_i - y} \exp(\theta_i y)}$$

   [[replace thetai with z expressions throughout??]] The target of inference μ_0 represents the ratio of the treatment to control log-odds.[[ref to bivriate model which models baseline risk, mentions assumptions in this approach]]

3. Single arm, incidence rate. Each study reports the number of events Y_i and the duration of observation T_i. The model is Poisson,

$$f(y_i \mid Z_i) = \exp(-\lambda)\frac{\lambda^{y_i}}{y_i!}$$
$$\lambda = T_i \exp(\mu_0 + \tau_0 Z_i)$$

   The target of inference represents the log of the incidence rate. The follow-up time T_i is treated as non-random.

4. Two arms, incidence rates. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the duration of observation $T_i^{(T)}$ and $T_i^{(C)}$. the model is binomial

$$f(y_i^{(T)} \mid Z_i) = \binom{m_i^{(T)}}{y_i^{(T)}} p^{y_i^{(T)}}(1-p)^{m_i^{(T)} - y_i^{(T)}}$$
$$p = \text{expit}(\log(T_i^{(T)}/T_i^{(C)}) + \mu_0 + \tau_0 Z_i)$$

   The target of inference is the logarithm of the ratio of the incidence rates between the two arms. As before the periods $T_i^{(T)}$ and $T_i^{(C)}$ are teated as non-random.

In all cases the likelihood for study i has the form:

$$L(\mu, \tau^2|Y_i) = \int_{-\infty}^{\infty} f(Y_i|z)\phi(z)dz$$

[[this is suppressing the dependnce of f on i]] where $f(y|z)$ is the conditional PMF given the random effect, as given above.

The likelihood ratio test of the null $\mu = \mu_0$ is based on the test statistic

$$W = 2\sum_{i=1}^{n}(\log L_i(\hat{\mu}, \hat{\tau}^2|Y_i) - \log L_i(\mu_0, \hat{\tau}_0^2|Y_i))$$

[[subscript on i to transmit dependence of pmf f on i]] where $(\hat{\mu}, \hat{\tau}^2)$ maximizes $(\mu, \tau^2) \mapsto \sum_{i=1}^{n} \log L_i(\mu, \tau^2|Y_i)$ and $\hat{\tau}_0^2 = \hat{\tau}_0^2(\mu)$ maximizes $\tau^2 \mapsto \sum_{i=1}^{n} \log L_i(\mu_0, \tau^2|Y_i)$. By Wilks theorem in routine applications W converges to a chi squared distribution with a single degree of freedom under the null hypothesis. Relying on this asymptotic approximation a confidence interval for μ_0 can be formed by inverting the test statistic.

Under a range of conditions[[, chiefly the existence of an edgeworth expansion for the underlying random variables,]] the mean of a correctly specified LR statistic ((ref)) admits an expansion in integral powers of $1/n$

$$\mathbb{E}(W) = 1 + b/n + O(1/n^2).$$

It then follows that

$$W_B = \mathbb{E}(\frac{W}{1 + b/n})$$

is bias-corrected to order $O(1/n^2)$. The work of ((lawley and bartlett)) shows further that, under appropriate conditions, not only does the mean of ((ref above)) match the mean of a chi squared to order $O(1/n^2)$, but also all other moments, and so the entire distribution. Then the resulting asymptotic rate of convergence to a chi squared random variable is $O(1/n^2)$. As with ((ref lr statistic)), the adjusted statistic may be inverted to obtain CIs and p-vals. [[mention C depends on parameters, or maybe make explicit in the notation]]

Among the assumptions underlying the above development we focus on two prominent ones. The first is correct specification of the likelihood, in particular, gaussianity of the unobserved random effect. we refer to sitjnen to other modeling assumptions implicit in ((ref 1–4 above)), such as ... . while the LR test itself depends on gaussianity for valid inference ((ref showing bad estimatio of tau2)), the bartlett correction doubles down on the assumption. second, the arguments for the bartlett correction to deliver 1/n2 rate generally depend on cramer's condition for the underlying RVs, which excludes lattice variables such as are considered here. open question since the 80s, when lattice RVs can be bartlett codnitions. cramers condition has been relaxed in recent work ((cite)) but these conditions are difficult to evaluate in specific cases. they do offer promise that the entireprise of bartlett correcting count data, which has been done since 80s ((cite cordeiro)), is not pointless. we make not theoretical contibutions here and will examine both model robustness and discreteness by simulation.

3. proposed method.

lawley ((cite)) provides expressions for the correction ((ref)) as sums and products of the components of the inverse fisher information and certain tensors [[k terms]], defined as follows. Let partial D_r denote partial differentiation with respect to a sequence of the parameters μ and τ^2 given by r. Define $k_r = \mathbb{E}D_r l(\mu, \tau^2)$ and $k_r^{(s)} = D_s k_r$. In this notation the Fisher information matrix has components k_{11}, k_{12}, k_{22}.[[intro component notation or use mu,tau2 here?]]. Lawley's formulas for computation of the correction require all terms k_r where r is of length 2, 3, or 4, and all terms $k_r^{(s)}$ where r is of length 3 and s length 1 or r is of length 2 and s of length 1 or 2. Taking into account permutation symmetry of the order of differentiation, there are 35 such terms.

To compute the tensor components ((ref)) for the likelihood ((ref)), we approximate the integral by a quadrature rule. since all the likelihoods involve integrating against a gaussian, gauss-hermite approximation is a natural choice, replacing the integral ((ref)) with a weighted sum. ((how much detail to give here, maybe just leave it as a sum approximation.))

$$L(\theta, \tau^2|Y) = \int_{-\infty}^{\infty} f(Y|z)\phi(z)dz \approx \sum_j w_j f(Y|z_j)$$

The higher-order partial derivative of the log-likelihood is approximated by $D_r \log L(\mu, \tau^2|Y) \approx D_r \log \sum_j w_j f(Y|z_j)$. The expectation of the latter may be approximated by another quadrature

$$\begin{aligned} k_r &= \sum_y \left(D_r \log L(\mu, \tau^2|y)\right) \int f(y|z)\phi(z)dz \\ &\approx \sum_y \left(D_r \log L(\mu, \tau^2|y)\right) \sum_k w_k f(y|z_k) \\ &\approx \sum_y \left(D_r \log \sum_j w_j f(y|z_j)\right) \sum_k w_k f(y|z_k) \end{aligned}$$

The outer sum is taken over the support of Y, which is a finite set in all the models considered here other than the Poisson model. The Poisson model requires a truncation rule, discussed further below. the general tensor component is then

$$k_r^{(s)} \approx \sum_y D_s \left(\left(D_r \log \sum_j w_j f(y|\theta_j)\right) \sum_k w_k f(y|\theta_k)\right)$$

Carrying these differentiations through yields rational expressions in derivatives of the weighted sums $\sum_j w_j f(y|z_j)$. The form of these 35 expressions, for the 35 k terms, are the same for any model based on a likelihood of the form ((ref)), and may be pre-computed. For example $k_{\mu,\mu} = F_{20} - (F_{10}^2/F_{00})$, and $k_{\mu,\tau^2}^{(\tau^2)} = (F_{01}^2 F_{10})/F_{00}^2 - (F_{02}F_{10} + F_{01}F_{11})/F_{00} + F_{12}$, letting $F_{jk} = d^j/d\mu^j d^k/d(\tau^2)^k \sum_j w_j f(y|\theta_j)$.

The derivatives of the model-specific conditional PMFs that appear in the 35 tensor components number 15: F_{jk} for $0 \leq j + k \leq 4$. The form of these derivatives may be pre-computed but their specific values in general depend on parameters of the conditional

PMF ((see above)), and must typically be computed for each study of a meta-analysis. using well-known recurrences the derivatives may be re-written as polynomial expressions of the conditional PMFs given in ((ref 1–4)), for which fast and numerically stable routines are available. For example: ... . [[see appendix for similar recurrences, or refer to johnson and kotz or wherever]]

further implementation details.

chain rule: differentiate wrt pmf sums. get 35 k terms in terms of pmfsums (give an example or two).

differentiate pmfs using recurrences in terms usual pmfs to allow fast, stable routines to be used.

some specific issues: support for poisson. speed issues with hyperg.

3. simulation

goals: robustness/gaussianity check. discreteness issue.

4. data analysis.

4a. proportions

4b. incidence